A neural network potential for water clusters

Lesson notes accompanying `tutorial_02_wclusters/wclusters.ipynb`

Machine-learned interatomic potentials — hands-on session

Abstract

The second notebook applies the workflow of the Zundel tutorial — committee neural networks, query by committee (QbC), MD in CP2K, validation by observables — to a protonated water cluster $(H_2O)_6H^+$. It then asks the question that matters most in practice: *how far does the model transfer?* It tests the model on a cluster almost three times larger, $(H_2O)_{16}H^+$, and on periodic bulk water. Finally, it questions the reference method itself. This is still a **basic, pedagogical example**: small clusters in vacuum, a semi-empirical reference and a tiny network. Its failures are as instructive as its successes. These notes assume you have done the Zundel tutorial and its lesson. The last section extends its discussion of the subtleties beyond the simple model with points specific to water.

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1 From one ion to a cluster

The Zundel ion had 7 atoms, all within a single cutoff sphere. Every atom "saw" the whole molecule, and the network effectively learned one global function. $(H_2O)_6H^+$ (19 atoms) is the first system where the *locality* of the Behler–Parrinello model really matters:

- The cluster is ~ 7.5 Å across, more than the cutoff $r_c = 12\,a_0 \approx 6.35$ Å, so atoms on opposite sides do not see each other.
- It contains a hydrogen-bond network. The excess proton can take Zundel-like ($H_5O_2^+$) and Eigen-like ($H_9O_4^+$) local configurations and hop between waters [1]. The model must describe both, from local information only.
- There are now many oxygens, so the O–O–O angular functions excluded in tutorial 1 are switched back on (`exclude_triples` is commented out). Both elements now have 32 descriptors and 201 weights.

Protonated water clusters are a classic test case for MLIPs. They have been fitted up to coupled-cluster accuracy with exactly this type of model and an automated, committee-based selection of training data [2].

What this example is — and what it is not

Training system	$(H_2O)_6H^+$, 19 atoms, vacuum.
Test systems	$(H_2O)_{16}H^+$ (49 atoms, vacuum); 64 H_2O in a periodic box (192 atoms, NpT).
Reference	SCC-DFTB [3]. Known to describe water–water interactions poorly (Sec. 6).
Model	Same as tutorial 1: 2 hidden layers × 5 neurons per element, 8-member committee.

A research model for aqueous systems would use a much better reference (hybrid DFT with dispersion, or coupled cluster), bigger or equivariant networks, and above all training data that include the condensed phase (Sec. 7).

2 Reference data

All reference trajectories in `data/` were generated with SCC-DFTB in CP2K, with a Nosé–Hoover chain thermostat at 300 K ($\tau = 50$ fs) and $\Delta t = 0.5$ fs:

directory	system	atoms	ensemble, length	role
`data/cluster`	$(H_2O)_6H^+$	19	NVT, 25 ps, 10 001 frames	training + validation
`data/big_cluster`	$(H_2O)_{16}H^+$	49	NVT, 25 ps, 10 001 frames	validation only
`data/bulk`	64 H_2O	192	NpT, 1 atm, 20 ps	density only (cell file)

The key point: **the model is trained only on the small cluster.** Everything else tests transferability.

Q. *Before running anything: which local environments occur in bulk water but never in a 7-water cluster? Think about the number of neighbours inside a 6.35 Å sphere, and about surface molecules.*

3 Active learning on the cluster

This notebook skips the naive single model and goes directly to QbC, with the same settings as tutorial 1: 20 random initial structures, then 15 iterations that each add the 20 structures (out of 500 random candidates) with the largest force disagreement, using 8 members and 15 epochs. The final committee is retrained for 50 epochs on the last training set.

Table 1: Mean committee force disagreement (meV/Å) during QbC on $(H_2O)_6H^+$ in a test run, and wall time per iteration (8 cores).

iteration	training-set size	train	candidates	selected	time (s)
1	20	129	151	228	12
2	40	100	102	142	13
3	60	79	75	98	13
5	100	62	56	73	14
10	193	50	43	51	16
15	281	45	38	49	19

Compare with the Zundel run (tutorial 1 lesson, Table 1):

- The initial disagreement is much lower (151 vs. 695 meV/Å on candidates). The cluster offers many more atoms per structure, and hence many more force labels, for the same 20 structures.
- The final disagreement is similar (~ 40 meV/Å), with a similar number of training structures ($\sim$280).
- This time the cost per iteration *grows*, from 12 to 19 s. Training cost scales with the number of atoms in the training set, and each structure now has 19 atoms instead of 7.
- Late iterations re-select a few structures already in the set ("n already present"), a sign that the pool is exhausted.

Q. *The disagreement converges to ~ 40 meV/Å, not to zero. What limits it: the data, the network size (10 hidden neurons), or the locality approximation? How could you tell them apart?*

4 MD and validation on the training system

The committee runs 10 ps of NVT MD on $(H_2O)_6H^+$ (`nnp_md/`). In the test run it stayed stable: mean temperature 297 K (reference 303 K), and no molecule left the cluster.

score (%)	$H_5O_2^+$ (tut. 1)	$(H_2O)_6H^+$	$(H_2O)_{16}H^+$
RDF O–O	92.8	87.4	66.3
RDF O–H	97.0	91.0	82.9
RDF H–H	97.5	94.8	77.5
VDOS (mean of O, H)	90.6	79.9	82.4*

Score $= 100(1-\varepsilon)$, with ε the normalised absolute deviation from the DFTB curve (tutorial 1 lesson, Sec. 6).
*Recomputed; see the warning box in Sec. 5.

The scores are lower than for the Zundel ion, even on the training system. There are three reasons. The system is more complex, but the network is the same size. The NNP run is shorter than the reference (10 vs. 25 ps), which adds statistical noise to the RDFs and, above all, to the spectra. And a cluster in vacuum is a "soft" object: its shape fluctuates and molecules can drift outwards.

Molecules flying away

The notebook warns that you may see a water molecule leave the cluster, and attributes it to DFTB, which *significantly* underestimates water–water interactions. This is not an NNP artefact: the DFTB reference does it too (see next section). An MLIP reproduces the physics of its reference, including the bad parts.

5 Transfer 1: a larger cluster, $(H_2O)_{16}H^+$

The same model now runs 10 ps on a 49-atom cluster it has never seen (`big_cluster/`), and is compared with a 25 ps DFTB reference. The run is stable (mean 302 K), but the RDF scores drop sharply, especially O–O (66 %).

Read the numbers critically. In the test run:

- The *DFTB reference* cluster loses water molecules. After 11.8 ps one oxygen sits more than 8 Å from the centre of mass, and by 25 ps three molecules have evaporated (up to 15.6 Å away).
- The NNP trajectory stops at 10 ps, *before* evaporation would have happened in the reference. One molecule made a temporary excursion beyond 8 Å at 2.2 ps and came back.

Part of the low O–O score therefore comes from comparing *different physical situations*: an intact cluster against one that is falling apart. It is not purely a model error. Comparisons by observables

are only meaningful for equivalent, converged simulations: same length, same ensemble, same state of the system.

Bug in the notebook: wrong VDOS for the big cluster

The last line of the big-cluster scoring cell reads

```
mlps.run_vdos_test(ref_atv, dt_trj, nnp_atv, dt_trj)
```

It reuses the velocities of the *small* cluster, so the printed VDOS score is just a copy of the previous one (79.8 / 80.0 %). The correct call is

```
mlps.run_vdos_test(ref_big_atv, dt_trj, nnp_big_atv, dt_trj)
```

which in the test run gave 87.3 % (O) and 77.5 % (H), a mean of 82.4 %.

Q. *(Notebook) The scores seem almost reasonable, but from a visual inspection of the trajectories, are the dynamics well reproduced? What do you conclude about accuracy criteria, and about transferability?*

6 Transfer 2: bulk water

The last test is the hardest: 64 neutral water molecules in a periodic box, NpT at 1 atm and 300 K, starting at $\rho = 1.0\,\mathrm{g/cm^3}$ (`bulk/`, 2.5 ps). Note the two changes at once: no excess proton, and no vacuum.

What happened in the test run. The NNP liquid *expands*. The density fell from 1.00 to $0.52\,\mathrm{g/cm^3}$ within 2.5 ps, while the temperature peaked at 1127 K. The model does not hold the liquid together. In bulk, each molecule has neighbours in every direction inside the cutoff, an environment entirely absent from a 7-water cluster, where every molecule sits at a surface. Concretely, an oxygen in the 7-water cluster has at most 6 other oxygens inside the cutoff; in the bulk box it has 28–38 (34 on average). The radial descriptors, which essentially count neighbours, take values far outside the training range. The network extrapolates, and the predicted forces (and hence the pressure) are meaningless.

And the reference? The notebook then plots the density of the DFTB bulk simulation (`data/bulk/bulk-1.cell`). It goes the other way: the box shrinks from 12.42 to 11.13 Å, and the density settles around $1.41\,\mathrm{g/cm^3}$, about 40 % above the experimental $0.997\,\mathrm{g/cm^3}$. This DFTB parametrisation produces a strongly over-structured, over-dense liquid. Even a perfect MLIP trained on it would be a perfect model of the wrong water.

Q. *(Notebook) What is happening in the NNP bulk run? Keep in mind that CP2K writes unwrapped coordinates, and that the viewer does not wrap atoms back into the periodic box, so molecules seem to "leave" the box even in a healthy simulation. Why is the density (cell file) a more reliable diagnostic here than the movie?*

Lessons of this tutorial

1. **Locality enables transfer, data limit it.** A local model can run on any system size, but it is only reliable for local environments represented in its training set.
2. **Validate with care.** Observable-based scores depend on how the comparison is set up (run length, state of the system), and the code that computes them can have bugs.
3. **The reference sets the ceiling.** Check the reference method against experiment or a higher level of theory *before* investing in the ML model.

7 Going further: subtleties beyond the simple model

The general refinements — equivariant message-passing networks, foundation models and fine-tuning, long-range electrostatics, energy-conserving benchmarks — are summarised in Sec. 8 of the Zundel lesson. Here are the points specific to this tutorial.

Clusters *can* teach the bulk, if done right. This tutorial shows a cluster-trained model failing in bulk. Yet one study did exactly the opposite, on purpose: it trained a Behler–Parrinello network on *finite clusters* computed with CCSD(T), the "gold standard" of quantum chemistry, which is unaffordable for periodic liquids. The model then ran coupled-cluster-quality MD of bulk liquid water, including nuclear quantum effects [6]. The difference: their clusters are large and chosen so that, within the cutoff, they contain bulk-like environments, and the selection is automated [2]. Transferability is a property of the *training set*, not of the model.

Better references are affordable. Because MLIPs are cheap, the reference can be expensive. Hybrid DFT with dispersion is the usual choice for aqueous systems [5], and coupled-cluster-level models of water exist [6]. A production water model would not be built on SCC-DFTB. In this tutorial it is a deliberate choice, to keep the data cheap and to show what a poor reference does.

Active learning in the loop. Here QbC only selects from an *existing* DFTB trajectory. In research workflows the committee runs the MD itself. Configurations with high disagreement (for example, the first frames of the bulk run above) are sent for new reference calculations and added to the training set, iteratively, until the disagreement stays low in production [4, 5]. Tutorial 3 uses models built this way for water at interfaces, in confinement and with ions.

Foundation models as a starting point. A pretrained universal model [7, 8] would not explode in bulk water the way this model does: it has seen condensed phases. But its accuracy for water depends on the reference functional of its training data, and charged clusters such as $(H_2O)_nH^+$ need a model trained on charged systems (e.g. on OMol25 [9]). A natural refinement is to fine-tune such a model on a few hundred structures selected by committee disagreement. That is the same idea as QbC, applied to a much better starting point.

Long-range electrostatics and the excess charge. A +1 cluster in vacuum has a net charge, and its surface molecules have a long-range electrostatic environment that a 6.35 Å cutoff cannot see. Charge-aware architectures [10, 11] address exactly this.

8 Exercises

1. Plot the training points selected by QbC on top of $E_p(t)$. Compare with the Zundel case.
2. Change the architecture (`n_hidden`, `n_neurons`). Does the final committee disagreement go below ~ 40 meV/Å? (Remember to delete the old `qbc/iteration-*`, `qbc/results.shelf.*` and `train/final_training` directories first, as the notebook says.)
3. Fix the VDOS line for the big cluster and compare the scores.
4. Make the big-cluster comparison fair: score the NNP run against only the first 10 ps of the reference (use the `stop` argument of `mlps.load_with_cell`). How much do the RDF scores change?
5. Monitor the cluster's integrity, e.g. the largest distance of an oxygen from the centre of mass, in the reference and NNP runs of both clusters.

6. Bulk water: in `bulk/bulk.inp`, the per-member energies are switched off (`&ENERGIES OFF` in the `&NNP/&PRINT` section; the same holds for the cluster inputs). Switch them on as in tutorial 1's `zundel.inp` (`&ENERGIES SILENT` with `&EACH MD 1`), rerun, and compute the committee energy spread along the trajectory. When does it blow up compared with the density? How would an active-learning loop use this?

7. Discuss: what training set would you need for a model of bulk water? Of the air–water interface?

9 Running times

Measured on an 8-core workstation (one core per committee member):

step	wall time
QbC (15 iterations, 8 members)	3.9 min
final training (8 members, 281 structures, 50 epochs)	48 s
$(H_2O)_6H^+$ MD (20 000 steps, 10 ps)	46 s
$(H_2O)_{16}H^+$ MD (20 000 steps, 10 ps)	1.6 min
bulk water NpT MD (5000 steps, 2.5 ps, 192 atoms)	6.1 min

The bulk run is the longest step. Start it as soon as the final model is trained, and do the big-cluster analysis while it runs.

References

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